

Chapter 4. Robot Dynamics and Control

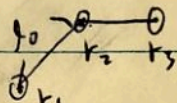
Motivation

Goal: Write the motion of the joints of robot by Lagrange's equation

$$F_i = m_i \ddot{r}_i \quad r_i: \text{position} \in \mathbb{R}^3, \quad i=1, \dots, n \quad (\text{not linked})$$

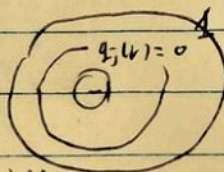
$$q_j(r_1, \dots, r_n) = 0 \quad j=1, \dots, k \quad k\text{-constraint}$$

ex) $\|r_1 - r_2\| = l_0$



$$F = \begin{bmatrix} m_1 I & & 0 \\ & \ddots & \\ 0 & & m_n I \end{bmatrix} \begin{bmatrix} \ddot{r}_1 \\ \vdots \\ \ddot{r}_n \end{bmatrix} + \sum_{j=1}^k \lambda_j \Gamma_j \quad (\text{Lagrange multiplier})$$

$$\Gamma_i \in \mathbb{R}^{3n} \quad \text{ex) } \Gamma_i = \nabla q_j$$



$3n + k$ equations with $3n + k$ variables.

• Computationally complex.

Consider $m = 3n - k$ variables $q_1, \dots, q_m$ (satisfying k -constraint)

$$r_i = f_i(q_1, \dots, q_m) \quad \Leftrightarrow \quad q_j(r_1, \dots, r_n) = 0$$

$$i=1, \dots, n \quad j=1, \dots, k$$

Def q_i : generalized coordinate.

$$L(q, \dot{q}) = \underbrace{T(q, \dot{q})}_{\text{kinetic E}} - \underbrace{V(q)}_{\text{Potential E}} \quad \text{Lagrangian}$$

Then Lagrange's equation

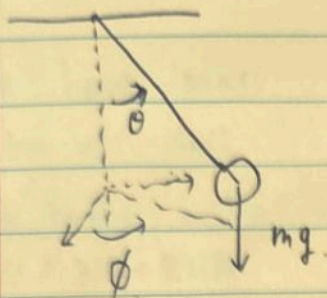
$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \Gamma_i \quad i=1, \dots, m$$

(minimizing path of L)
 $q(u) + \epsilon q(u)$

$$\bullet \quad T = \frac{1}{2} \sum m_i \|\dot{r}_i\|^2 \quad \Updownarrow$$

• without constraint $F_i = m_i \ddot{r}_i$

Ex 4.1 Spherical pendulum.



generalized coordinate : $q = (\theta, \phi)$ (original $v: 3$ constraint 1.)

$$r(\theta, \phi) = \begin{bmatrix} l \sin \theta \cos \phi \\ l \sin \theta \sin \phi \\ -l \cos \theta \end{bmatrix}$$

$$\dot{r}(\theta, \phi) = \begin{bmatrix} l \cos \theta \cos \phi \dot{\theta} - l \sin \theta \sin \phi \dot{\phi} \\ l \cos \theta \sin \phi \dot{\theta} + l \sin \theta \cos \phi \dot{\phi} \\ l \sin \theta \dot{\theta} \end{bmatrix}$$

Kinetic energy

$$T = \frac{1}{2} m \|\dot{r}\|^2 = \frac{1}{2} m l^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$$

Potential energy

$$V = -mgl \cos \theta$$

Lagrangian

$$L(q, \dot{q}) = \frac{1}{2} m l^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + mgl \cos \theta$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = \frac{d}{dt} (m l^2 \dot{\theta}) = m l^2 \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = m l^2 \sin \theta \cos \theta \dot{\phi}^2 - mgl \sin \theta$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} = \frac{d}{dt} (m l^2 \sin^2 \theta \dot{\phi}) = m l^2 2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + m l^2 \sin^2 \theta \ddot{\phi}$$

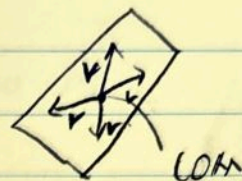
$$\frac{\partial L}{\partial \phi} = 0$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \begin{bmatrix} m l^2 & 0 \\ 0 & m l^2 \sin^2 \theta \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} + \begin{bmatrix} -m l \sin \theta \cos \theta \dot{\phi}^2 \\ 2 m l^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} \end{bmatrix} + \begin{bmatrix} mgl \sin \theta \\ 0 \end{bmatrix} = 0$$

2.2 Inertial properties of rigid bodies.

• Link have mass. $m = \int_V \rho(r) dV$

• Center of mass $\bar{r} = \frac{1}{m} \int_V \rho(r) r dV$



★ Body frame: Fix center of mass at $(0, 0, 0)$

• $x(t) = p(t) + R(t)r$ $\dot{x} = \dot{p} + \dot{R}r$

Kinetic E. $T = \frac{1}{2} \int_V \rho(r) \|\dot{p} + \dot{R}r\|^2 dV$

$$= \frac{1}{2} \int_V \rho(r) (\|\dot{p}\|^2 + \underbrace{2\dot{p}^T \dot{R}r}_{(ii)} + \underbrace{\|\dot{R}r\|^2}_{(iii)}) dV$$

$$(ii) \quad \dot{p}^T \dot{R} \int_V \rho(r) r dV = 0 \quad (COM = 0)$$

$$\begin{aligned} (iii) \quad \frac{1}{2} \int_V \rho(r) (\dot{R}r)^T (\dot{R}r) dV &= \frac{1}{2} \int_V \rho(r) (R\hat{w}r)^T (R\hat{w}r) dV \quad (R = R\hat{w}) \\ &= \frac{1}{2} \int_V \rho(r) (\hat{w}r)^T R^T R (\hat{w}r) dV \quad \left(\begin{array}{l} \hat{w}r = w \times r \\ = -r \times w \\ = -\hat{r}w \end{array} \right) \\ &= \frac{1}{2} \int_V \rho(r) (-\hat{r}w)^T (-\hat{r}w) dV \\ &= \frac{1}{2} w^T \int_V \rho(r) \hat{r}^T \hat{r} dV w \\ &:= \frac{1}{2} w^T \underbrace{I}_{\text{Inertia tensor}} w \end{aligned}$$

$$\hat{r}^T \hat{r} = rr^T - \|r\|^2 I$$

$$I_{xx} = \int_V \rho(r) (y^2 + z^2) dr$$

$$I_{xy} = - \int_V \rho(r) xy dr$$

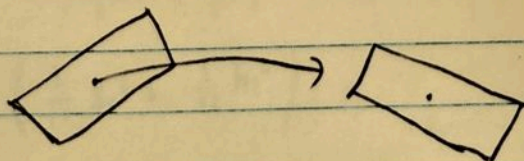
(Inertia tensor)

$$w = (w_1, w_2, w_3)$$

$$\hat{w} = \begin{pmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{pmatrix}$$

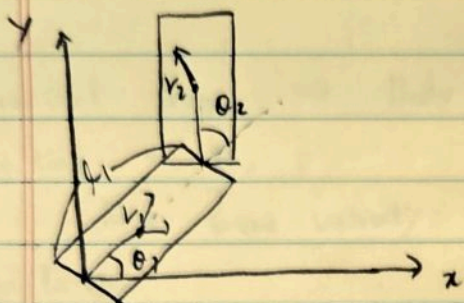
$$T = \frac{1}{2} m \|\dot{p}\|^2 + \frac{1}{2} w^T I w$$

* Important: V : Fixed body frame even V is moving.



$$\int_V dV = \int_a^b \int_c^d dx dy$$

Ex) Two link planar robot



rectangular bar with mass m_j
(equally distributed)

corn $(x_1, y_1) = (l_1 \cos \theta_1, l_1 \sin \theta_1)$

$$(x_2, y_2) = (l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2), l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2))$$

$$v_1 = (\dot{x}_1, \dot{y}_1) = (-l_1 \sin \theta_1 \dot{\theta}_1, l_1 \cos \theta_1 \dot{\theta}_1)$$

$$v_2 = (\dot{x}_2, \dot{y}_2) = (-l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2), l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos(\theta_1 + \theta_2) (\dot{\theta}_1 + \dot{\theta}_2))$$

$$w_1 = (0, 0, \dot{\theta}_1) \quad w_2 = (0, 0, \dot{\theta}_1 + \dot{\theta}_2)$$

$$T(\theta, \dot{\theta}) = \frac{1}{2} m_1 \|v_1\|^2 + \frac{1}{2} w_1^T I_1 w_1 + \frac{1}{2} m_2 \|v_2\|^2 + \frac{1}{2} w_2^T I_2 w_2$$

$$= \frac{1}{2} m_1 \|\dot{x}_1 + \dot{y}_1\|^2 + \frac{1}{2} (I_1)_{33} \dot{\theta}_1^2 + \frac{1}{2} m_2 \|\dot{x}_2 + \dot{y}_2\|^2 + \frac{1}{2} (I_2)_{33} (\dot{\theta}_1 + \dot{\theta}_2)^2$$

$$= \frac{1}{2} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}^T \begin{bmatrix} \quad \quad \quad \\ \quad \quad \quad \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

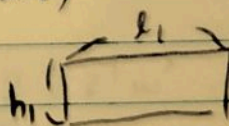
$$= L(\theta, \dot{\theta}) \Rightarrow \text{we can have Lagrange's equation}$$

$$(I_1)_{33} = \int_{V_1} \rho(r_1) (x_1^2 + y_1^2) dV_1 \quad (\text{Body frame})$$

$$= \int_{V_1} \rho(r_1) r_1^2 dV_1$$

$$= \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{m_1}{l_1 h_1} (x_1^2 + y_1^2) dx_1 dy_1$$

$$= m_1 \left(\frac{1}{12} l_1^2 + \frac{1}{12} h_1^2 \right)$$



$$\rho(r) = \frac{m_1}{l_1 h_1}$$

2.4 Newton - Euler equations.

Inertial frame $\Rightarrow$ Body frame.

$$\text{inertial } \begin{matrix} \underline{f}, \underline{\tau}, \dot{\underline{p}} \\ \text{force Torque Velocity} \end{matrix} \Rightarrow \text{Body } \begin{matrix} \underline{f}^b = R^T \underline{f} \\ \underline{v}^b = R^T \dot{\underline{p}} \end{matrix}$$

• Translation

$$\left(\begin{aligned} \underline{f} &= \frac{d}{dt}(m \dot{\underline{p}}) = \frac{d}{dt}(m R \underline{v}^b) = R m \dot{\underline{v}}^b + \dot{R} m \underline{v}^b \\ &= R m \dot{\underline{v}}^b + R \hat{\underline{\omega}}^b m \underline{v}^b \end{aligned} \right) \times R^T$$

$$\underline{f}^b = m \dot{\underline{v}}^b + \underline{\omega}^b \times (m \underline{v}^b)$$

• Rotation

$$\underline{I}' = R \underline{I} R^T : \text{ inertia tensor of inertial frame. } \left(\begin{array}{l} \underline{I}' : \text{ time depend} \\ \underline{I} : \text{ constant} \end{array} \right)$$

$$\underline{\tau} = \frac{d}{dt}(\underline{I}' \underline{\omega}^s)$$

$$= \underline{I}' \dot{\underline{\omega}}^s + \dot{\underline{I}}' \underline{\omega}^s$$

$$= \underline{I}' \dot{\underline{\omega}}^s + (\dot{R} \underline{I} R^T + R \underline{I} \dot{R}^T) \underline{\omega}^s \quad (\dot{R} = \hat{\underline{\omega}}^s R)$$

$$= \underline{I}' \dot{\underline{\omega}}^s + (\hat{\underline{\omega}}^s R \underline{I} R^T + R \underline{I} (R^T (-\hat{\underline{\omega}}^s))) \underline{\omega}^s$$

$$= \underline{I}' \dot{\underline{\omega}}^s + \hat{\underline{\omega}}^s \underline{I}' \underline{\omega}^s - \underline{I}' \hat{\underline{\omega}}^s \underline{\omega}^s \quad (\text{Euler's equation})$$

$$\underline{\omega}^b = R^T \underline{\omega}^s, \quad \underline{\tau}^b = R^T \underline{\tau}$$

$$\Rightarrow \begin{bmatrix} m \underline{I} & 0 \\ 0 & \underline{I} \end{bmatrix} \begin{bmatrix} \dot{\underline{v}}^b \\ \dot{\underline{\omega}}^b \end{bmatrix} + \begin{bmatrix} \underline{\omega}^b \times (m \underline{v}^b) \\ \underline{\omega}^b \times (\underline{I} \underline{\omega}^b) \end{bmatrix} = \begin{bmatrix} \underline{f}^b \\ \underline{\tau}^b \end{bmatrix}$$

(Newton - Euler equation)

$$\left(\begin{array}{c} R^T \hat{\underline{\omega}}^s R \underline{I} R^T \underline{\omega}^s \\ \parallel \\ R^T \underline{\omega}^s \times R = \widehat{R \underline{\omega}^s} = \hat{\underline{\omega}}^b \end{array} \right)$$

4.3. Dynamics of open chain Manipulator

4.3.1 Lagrangian for open chain robot

Let $g_{s_{i-1}}(0)$: COM of i th link

$$g_{s_i}(\theta) = e^{\hat{s}_1 \theta_1} \dots e^{\hat{s}_{i-1} \theta_{i-1}} g_{s_{i-1}}(0)$$

Body velocity of COM of i th link

$$V_{s_i}^b = J_{s_i}^b(\theta) \dot{\theta}$$

$$J_{s_i}^b(\theta) = [\zeta_1^+, \dots, \zeta_i^+ \ 0 \ \dots \ 0]$$

$$\zeta_i^+ = \text{Ad}^{-1}_{(e^{\hat{s}_1 \theta_1}, \dots, e^{\hat{s}_{i-1} \theta_{i-1}} g_{s_{i-1}}(0))} \zeta_i \quad (i \leq n)$$

Kinetic E of i th link

$$T_i(\theta, \dot{\theta}) = \frac{1}{2} (V_{s_i}^b)^T M_i V_{s_i}^b = \frac{1}{2} \dot{\theta}^T J_i^T(\theta) M_i J_i(\theta) \dot{\theta}$$

Total kinetic E

$$T(\theta, \dot{\theta}) = \sum_{i=1}^n T_i(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T \underline{M}(\theta) \dot{\theta} \quad (\text{different } M)$$

$$M(\theta) := \sum_{i=1}^n J_i^T(\theta) M_i J_i(\theta)$$

Potential E

$$V(\theta) = \sum_{i=1}^n m_i g h_i(\theta)$$

Lagrangian

$$L(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} - V(\theta)$$

Note

$$\begin{aligned} g \in SE(3) &= \left\{ g = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \mid \begin{array}{l} R \in SO(3) \\ p \in \mathbb{R}^3 \end{array} \right\} \\ \text{Ad}_g &= \begin{bmatrix} R & 0 \\ \hat{p}R & R \end{bmatrix} \\ \zeta &= \begin{pmatrix} v \\ \omega \end{pmatrix} \\ \text{Ad}_g \zeta &= \begin{bmatrix} Rv \\ \hat{p}Rv + R\omega \end{bmatrix} \end{aligned}$$

4.3.2. Equation of motion

Write $L(\theta, \dot{\theta}) = \frac{1}{2} \sum_{i,j} M_{ij}(\theta) \dot{\theta}_i \dot{\theta}_j - V(\theta)$

Lagrange's eqn

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = \tau_i$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} = \frac{d}{dt} \left(\sum_j M_{ij}(\theta) \dot{\theta}_j \right) = \sum_j \left(\underbrace{M_{ij}}_{||} \ddot{\theta}_j + \dot{M}_{ij} \dot{\theta}_j \right)$$

$$\frac{\partial L}{\partial \theta_i} = \frac{1}{2} \sum_{j,k} \frac{\partial M_{kj}}{\partial \theta_i} \dot{\theta}_k \dot{\theta}_j - \frac{\partial V}{\partial \theta_i} \quad \sum_{j,k} \frac{\partial \theta_k}{\partial t} \frac{\partial M_{ij}}{\partial \theta_k} \dot{\theta}_j$$

$$\Rightarrow \sum_j M_{ij} \ddot{\theta}_j + \sum_{j,k} \left(\frac{\partial M_{ij}}{\partial \theta_k} \dot{\theta}_j \dot{\theta}_k - \frac{1}{2} \frac{\partial M_{kj}}{\partial \theta_i} \dot{\theta}_k \dot{\theta}_j \right) + \frac{\partial V}{\partial \theta_i} = \tau_i$$

$$\sum_{j,k} \left(\frac{1}{2} \frac{\partial M_{ij}}{\partial \theta_k} \dot{\theta}_j \dot{\theta}_k + \frac{1}{2} \frac{\partial M_{jk}}{\partial \theta_i} \dot{\theta}_j \dot{\theta}_k \right) \quad \text{symmetric } j \leftrightarrow k$$

$$T_{ijk} := \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial \theta_k} + \frac{\partial M_{jk}}{\partial \theta_i} - \frac{\partial M_{ki}}{\partial \theta_j} \right) \quad (\text{Christoffel symbol})$$

$\dot{\theta}_i \dot{\theta}_j$: Coriolis force.

$\dot{\theta}_i^2$: centrifugal force.

Notation $C_{ij}(\theta, \dot{\theta}) = \sum_{k=1}^n T_{ijk} \dot{\theta}_k$ (Coriolis matrix)

Let $\tau_i = \tau_i - N_i(\theta, \dot{\theta})$

torque conservative force & friction.

$$\Rightarrow M(\theta) \ddot{\theta} + (C(\theta, \dot{\theta}) \dot{\theta} + N(\theta, \dot{\theta})) = \tau$$

* $M(\theta)$: symmetric positive definite

$N-ZC$: skew symmetric